

Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

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Time on Task:

7 hours, 29 minutes

Progress:

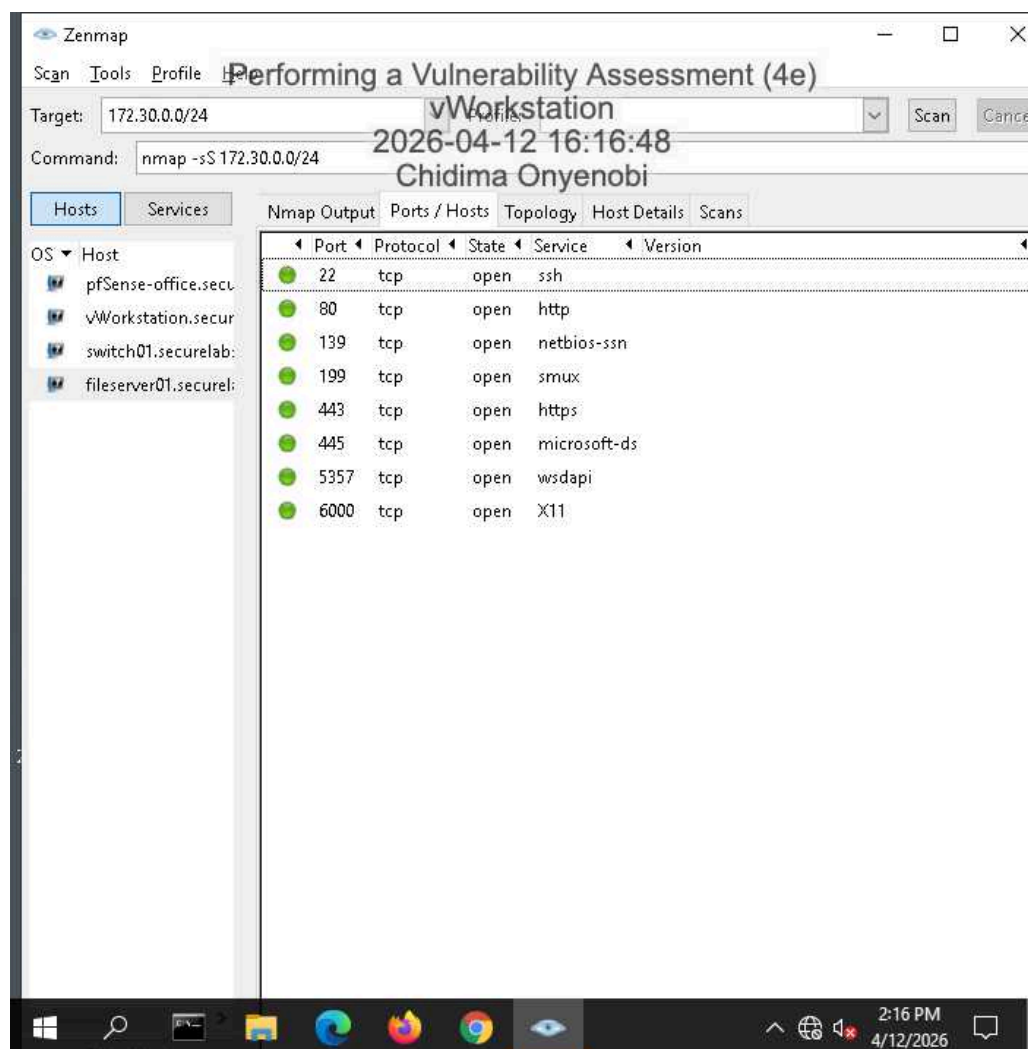
100%

Report Generated: Monday, April 13, 2026 at 12:37 AM

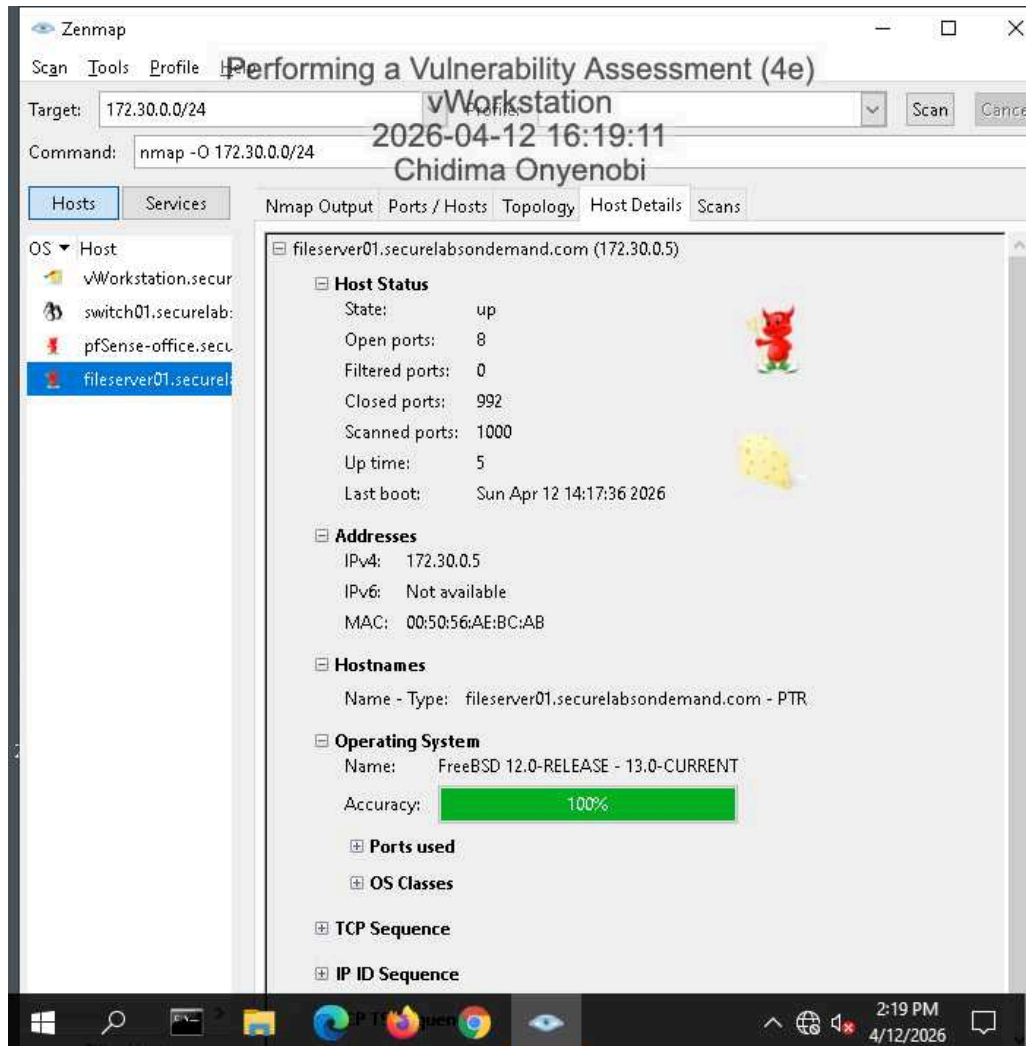
Section 1: Hands-On Demonstration

Part 1: Scan the Network with Zenmap

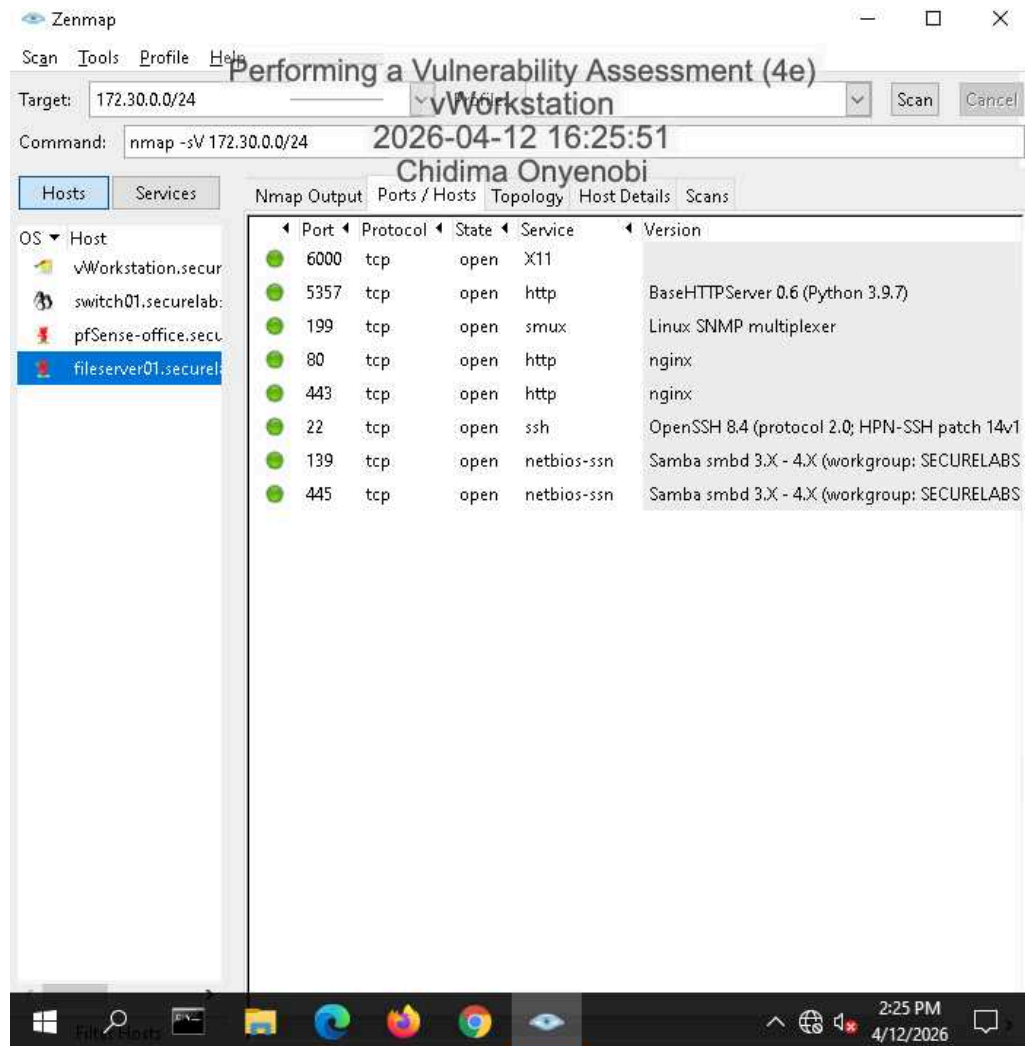
9. Make a screen capture showing the contents of the **Ports/Hosts** tab from the **SYN** scan for **fileserver01.securelabsondemand.com**.



15. Make a screen capture showing the contents of the **Host Details** tab from the OS scan for **fileserver01.securelabsondemand.com**.

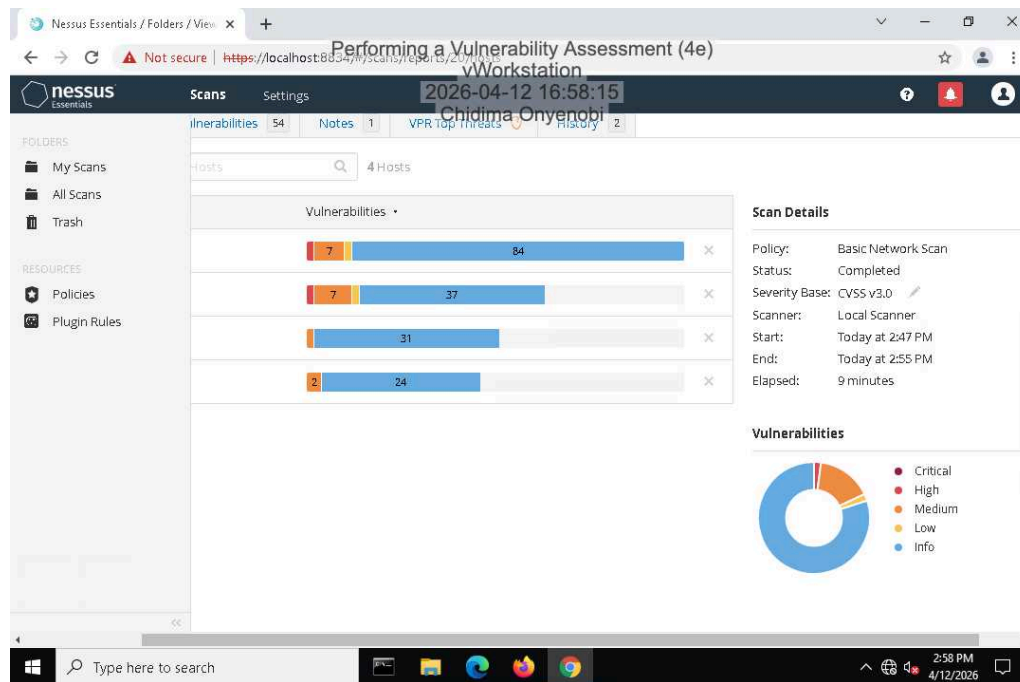


19. Make a screen capture showing the details in the **Ports/Hosts** tab from the **Service scan** for **fileserver01.securelabsondemand.com**.



Part 2: Conduct a Vulnerability Scan with Nessus

14. Make a screen capture showing the Nessus report summary.



Part 3: Evaluate Your Findings

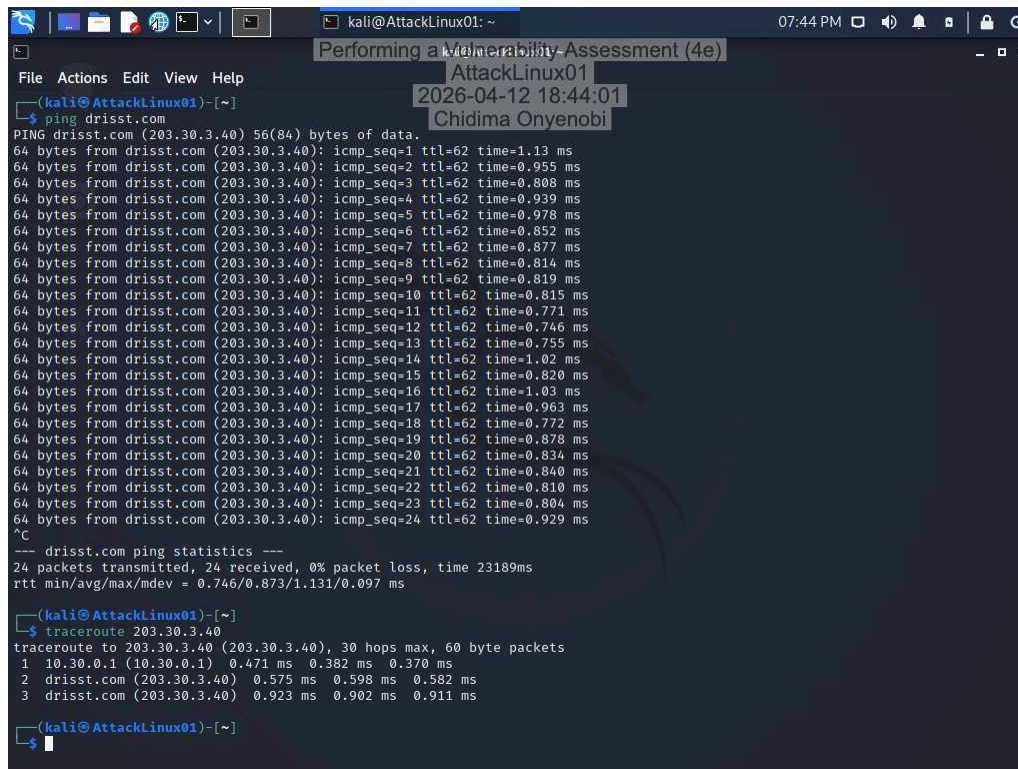
11. Summarize the vulnerability you selected, including the CVSS risk score, and recommend a mitigation strategy.

The vulnerability I will select is from 172.30.0.5. The "SSL Certificate can not be trusted" with a plugin ID of 51192. CVSS v3.0 of 6.5. A way you could migrate this might be to replace the certificate with a trusted certificate.

Section 2: Applied Learning

Part 1: Scan the Network with Nmap

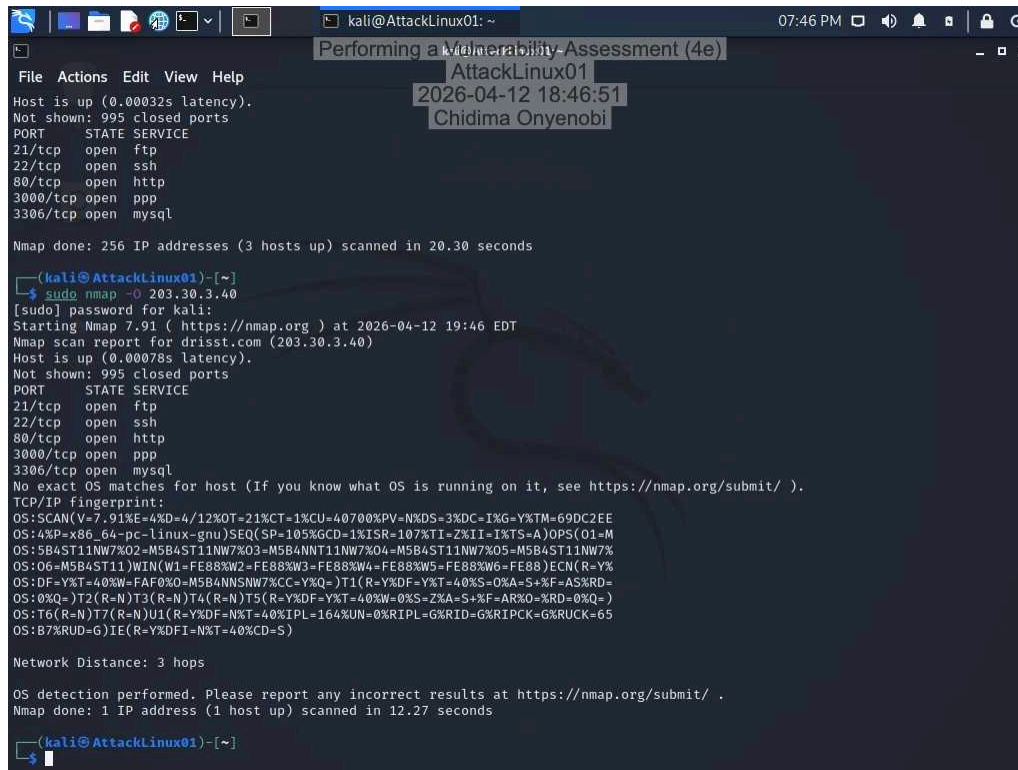
6. Make a screen capture showing the results of the traceroute command.



The screenshot shows a Kali Linux terminal window with the title bar "Performing a Vulnerability Assessment (4e)". The terminal output is as follows:

```
kali@AttackLinux01: ~  
$ ping drisst.com  
PING drisst.com (203.30.3.40) 56(84) bytes of data:  
64 bytes from drisst.com (203.30.3.40): icmp_seq=1 ttl=62 time=1.13 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=2 ttl=62 time=0.955 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=3 ttl=62 time=0.808 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=4 ttl=62 time=0.939 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=5 ttl=62 time=0.978 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=6 ttl=62 time=0.852 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=7 ttl=62 time=0.877 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=8 ttl=62 time=0.814 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=9 ttl=62 time=0.819 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=10 ttl=62 time=0.815 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=11 ttl=62 time=0.771 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=12 ttl=62 time=0.746 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=13 ttl=62 time=0.755 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=14 ttl=62 time=1.02 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=15 ttl=62 time=0.820 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=16 ttl=62 time=1.03 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=17 ttl=62 time=0.963 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=18 ttl=62 time=0.772 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=19 ttl=62 time=0.878 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=20 ttl=62 time=0.834 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=21 ttl=62 time=0.840 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=22 ttl=62 time=0.810 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=23 ttl=62 time=0.804 ms  
64 bytes from drisst.com (203.30.3.40): icmp_seq=24 ttl=62 time=0.929 ms  
^C  
--- drisst.com ping statistics ---  
24 packets transmitted, 24 received, 0% packet loss, time 23189ms  
rtt min/avg/max/mdev = 0.746/0.873/1.131/0.097 ms  
  
$ traceroute 203.30.3.40  
traceroute to 203.30.3.40 (203.30.3.40), 30 hops max, 60 byte packets  
1 10.30.0.1 (10.30.0.1) 0.471 ms 0.382 ms 0.370 ms  
2 drisst.com (203.30.3.40) 0.575 ms 0.598 ms 0.582 ms  
3 drisst.com (203.30.3.40) 0.923 ms 0.902 ms 0.911 ms
```

10. Make a screen capture showing the results of the Nmap scan with OS detection activated.



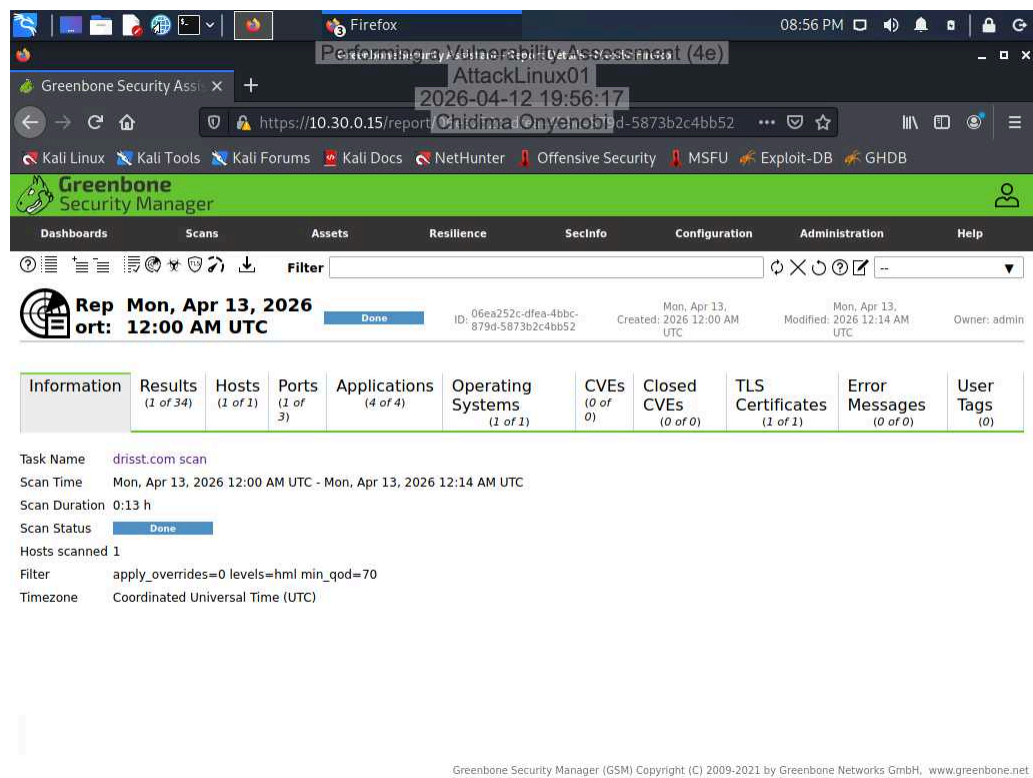
The screenshot shows a Kali Linux terminal window with the following content:

```
kali@AttackLinux01: ~  
Performing a Vulnerability Assessment (4e)  
AttackLinux01  
2026-04-12 18:46:51  
Chidima Onyenobi
```

```
File Actions Edit View Help  
Host is up (0.00032s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
  
Nmap done: 256 IP addresses (3 hosts up) scanned in 20.30 seconds  
  
(kali@AttackLinux01)-[~]  
$ sudo nmap -O 203.30.3.40  
[sudo] password for kali:  
Starting Nmap 7.91 ( https://nmap.org ) at 2026-04-12 19:46 EDT  
Nmap scan report for drisst.com (203.30.3.40)  
Host is up (0.00078s latency).  
Not shown: 995 closed ports  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
3000/tcp  open  ppp  
3306/tcp  open  mysql  
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/).  
TCP/IP fingerprint:  
OS:SCAN(V=7.91%E=4%D=4/12%OT=21%CT=1%CU=40700%PV=N%DS=3%DC=IXG-Y%TM=69DC2EE  
OS:4%P=x86_64-pe-linux-gnu)SEQ(SP=105%GCD=1%ISR=107%TI=Z%II=I%TS=A)OPS(O1=M  
OS:SB4ST11NW7%O2=M5B4ST11NW7%O3=M5B4NMT11NW7%O4=M5B4ST11NW7%O5=M5B4ST11NW7%  
OS:O6=M5B4ST11)WIN(W1=FE88%W2=FE88%W3=FE88%W4=FE88%W5=FE88%W6=FE88)ECN(R=Y%  
OS:DF=Y%T=40%W=FAF0%O=M5B4NNSNW7%CC=Y%Q=)T1(R=Y%DF=Y%T=40%W=0%S=0%KA=S%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=N)T5(R=Y%DF=Y%T=40%W=0%S=0%KA=S%F=AR%O=0%RD=0%Q=)T6(R=N)T7(R=N)U1(R=Y%DF=N%T=40%IPL=164%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=65OS:B7%RUD=G)IE(R=Y%DFI=N%T=40%CD=S)  
  
Network Distance: 3 hops  
  
OS detection performed. Please report any incorrect results at https://nmap.org/submit/.  
Nmap done: 1 IP address (1 host up) scanned in 12.27 seconds  
  
(kali@AttackLinux01)-[~]  
$
```

Part 2: Conduct a Vulnerability Scan with OpenVAS

13. Make a screen capture showing the detailed OpenVAS scan results.



Part 3: Prepare a Penetration Test Report

Target

Insert the target here.

drisst.com (203.30.3.40)

Completed by

Insert your name here.

Gift Onyenobi

On

Insert current date here.

April 12, 2026

Purpose

Identify the purpose of the penetration test.

The purpose of the penetration test is to identify the vulnerability in the drisst.com web server.

Scope

Identify the scope of the penetration test.

The scope for this test is to identify the vulnerability and comply with security standards to personalize recommendations for this program.

Summary of Findings

Identify and summarize each of the three high-severity vulnerabilities identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

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Conclusion

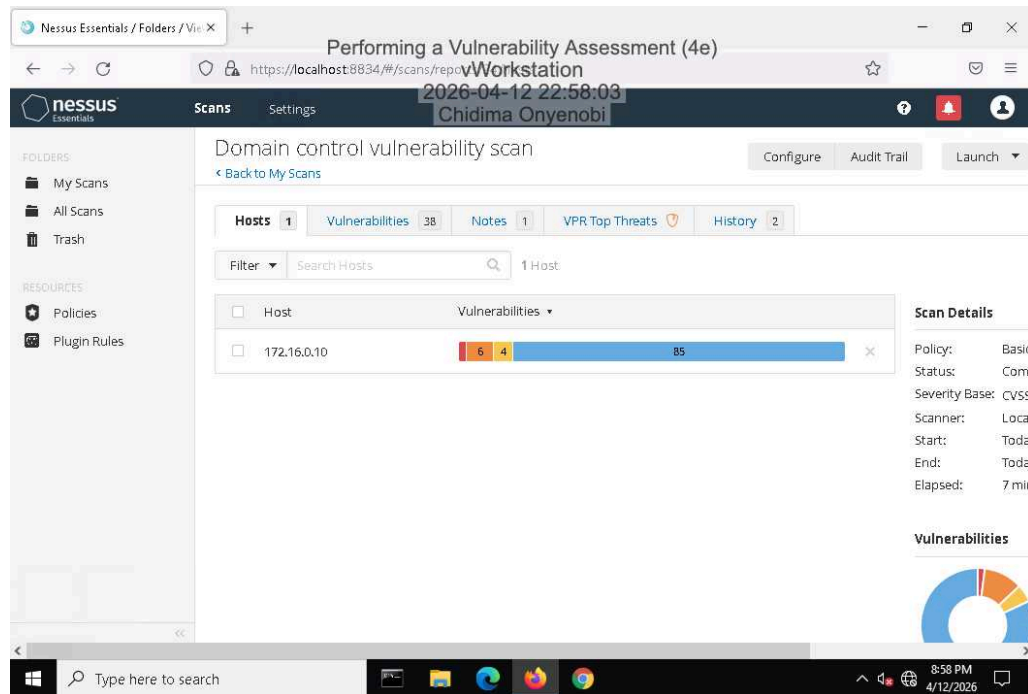
Identify your key findings.

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Performing a Vulnerability Assessment (4e)

Fundamentals of Information Systems Security, Fourth Edition - Lab 02

Make a screen capture showing the **Nessus report summary** for the domain controller.



Part 3: Prepare a Penetration Test Report

Target

Insert the target here.

Domain control

Completed by

Insert your name here.

Gift Onyenobi

On

Insert current date here.

April 12, 2026

Purpose

Identify the purpose of the penetration test.

The purpose is to analyze security vulnerabilities within the target's environment.

Scope

Identify the scope of the penetration test.

The scope is limited to 172.16.0.10

Summary of Findings

Identify and summarize each vulnerability identified during your penetration test. For each vulnerability, identify the severity, describe the issue, and recommend a remediation.

It has a total of 95 vulnerabilities. A high finding, little to no critical, 6 medium vulnerabilities, 4 low, and 85 info. Please prioritize patching the high, medium, and low vulnerabilities and schedule maintenance at your earliest convenience.

Conclusion

Identify your key findings.

An outdated system and unpatched services